Assignment of day - 6

1. Given a python list. Write a program to add all its elements into a given set.

```
[1]: sample_set = {"Yellow", "Orange", "Black"}
     sample_list = ["Blue", "Green", "Red"]
     sample_set.update(sample_list)
     print(sample_set)
```

```
{'Blue', 'Red', 'Black', 'Orange', 'Green', 'Yellow'}
```

2.Return a new set of identical items from two sets.

```
[2]: set1 = {10, 20, 30, 40, 50}
     set2 = {30, 40, 50, 60, 70}
     print(set1.intersection(set2))
```

```
{40, 50, 30}
```

3. Write a python program to return a new set with unique items from both sets by removing duplicates.

```
[3]: set1 = {10, 20, 30, 40, 50}
     set2 = {30, 40, 50, 60, 70}
     print(set1.union(set2))
```

```
{70, 40, 10, 50, 20, 60, 30}
```

4. Given two python sets. Write a python program to update the first set with items that exist only in the first set and not in the second set.

```
[4]: set1 = {10, 20, 30}
     set2 = {20, 40, 50}
     set1.difference_update(set2)
     print("set1", set1)
```

```
set1 {10, 30}
```

5. Write a program to remove items 10, 20, 30 from the following set at once.

```
[5]: set1 = {10, 20, 30, 40, 50}
     set1.difference_update({10, 20, 30})
     print(set1)
```

```
{50, 40}
```

6. Return a set of elements present in set A or B but not both.

```
[6]: set1 = {10, 20, 30, 40, 50}
     set2 = {30, 40, 50, 60, 70}
     print(set1.symmetric_difference(set2))
```

```
{20, 70, 10, 60}
```

7. Check if the two sets have any elements in common. If yes, display the common elements.

```
[7]: set1 = {10, 20, 30, 40, 50}
     set2 = {60, 70, 80, 90, 10}
     common = set1.intersection(set2)
     if common:
         print("Two sets have items in common")
         print(common)
```

```
Two sets have items in common
{10}
```

8. Update set - 1 by adding items from set – 2, except common items.

```
[8]: set1 = {10, 20, 30, 40, 50}
     set2 = {30, 40, 50, 60, 70}
     set1 = set1.symmetric_difference(set2)
     print(set1)
```

```
{20, 70, 10, 60}
```

9. Remove items from set – 1 that are not common to both set – 1 and set – 2.

```
[9]: set1 = {10, 20, 30, 40, 50}
     set2 = {30, 40, 50, 60, 70}
     set1.intersection_update(set2)
     print(set1)
```

```
{40, 50, 30}
```

10. Reverse the tuple.

```
[10]: tuple1 = (10, 20, 30, 40, 50)
      print(tuple1[::-1])
```

```
(50, 40, 30, 20, 10)
```

11. The given tuple is a nested tuple. Write a python program to print the value 20.

```
[11]: tuple1 = ("Orange", [10, 20, 30], (5, 15, 25))
      print(tuple1[1][1])
```

```
20
```

12. Write a program to unpack the following tuple into four variables and display each variable.

[12]:
```
tuple1 = (10, 20, 30, 40)
a, b, c, d = tuple1
print(a)
print(b)
print(c)
print(d)
```

```
10
20
30
40
```

13. Swap two tuples in python.

[13]:
```
tuple1 = (11, 22)
tuple2 = (99, 88)
tuple1, tuple2 = tuple2, tuple1
print("tuple1:", tuple1)
print("tuple2:", tuple2)
```

```
tuple1: (99, 88)
tuple2: (11, 22)
```

14. Write a program to copy elements 44 and 55 from the following tuple into a new tuple.

[14]:
```
tuple1 = (11, 22, 33, 44, 55, 66)
tuple2 = tuple1[3:5]
print("tuple2:", tuple2)
```

```
tuple2: (44, 55)
```

15. Given is a nested tuple. Write a program to modify the first item (22) of a list inside a following tuple to 222.

```
tuple1 = (11, [22, 33], 44, 55)
tuple1[1][0] = 222
print("tuple1:", tuple1)
```

```
tuple1: (11, [222, 33], 44, 55)
```

16. Sort a tuple of tuples by 2nd item.

[16]:
```
tuple1 = (('a', 23), ('b', 37), ('c', 11), ('d', 29))
sorted_tuple = tuple(sorted(tuple1, key=lambda x: x[1]))
print(sorted_tuple)
```

```
(('c', 11), ('a', 23), ('d', 29), ('b', 37))
```

17. Count the no.s of occurrence of item 50 from a tuple.

[17]:
```
tuple1 = (50, 10, 60, 70, 50)
print(tuple1.count(50))
```

```
2
```

18. Check if all items in the tuple are the same.

[18]:
```
tuple1 = (45, 45, 45, 45)
print(all(x == tuple1 for x in tuple1))
```

```
False
```